

CHE 201 Spring 2020 HW 6

Thanh Nguyen

TOTAL POINTS

100 / 100

QUESTION 1

1 Question 1 15 / 15

✓ - 0 pts Correct

- 1 pts Wrong unit
- 9 pts Wrong volume of air per kilogram
- 6 pts Wrong understanding of volumetric flow rate

- 5 pts Not enough evaluation on economic factors
- 2.5 pts Not enough evaluation on the effects on the ecosystem
- 5 pts Not enough evaluation on the effects on the ecosystem
- 15 pts Not answered

QUESTION 2

2 Question 2 15 / 15

✓ - 0 pts Correct

QUESTION 3

3 Question 3 15 / 15

✓ - 0 pts Correct

- 2.5 pts a) Wrong calculation
- 2.5 pts a) Wrong equation
- 2.5 pts b) Wrong calculation
- 2.5 pts b) Wrong equation
- 2.5 pts c) Wrong calculation
- 2.5 pts c) Wrong equation

QUESTION 4

4 Question 4 20 / 20

✓ - 0 pts Correct

QUESTION 5

5 Question 5 20 / 20

✓ - 0 pts Correct

QUESTION 6

6 Question 6 15 / 15

✓ - 0 pts Correct

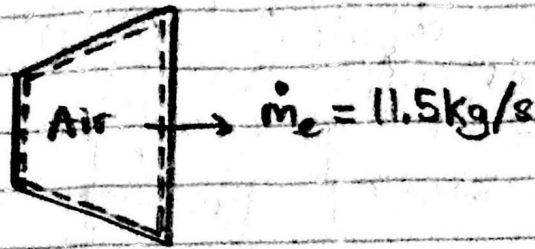
- 2.5 pts Not enough evaluation on technical factors
- 5 pts Not enough evaluation on technical factors
- 2.5 pts Not enough evaluation on economic factors

Thanh Nguyen

CHE 201 - HW6

1.) Sketch

$$p = 200 \text{ kPa}$$
$$T = 150^\circ\text{C}$$



Engineering Model

1. The system is a control volume
2. Ideal gas behavior

Analysis

$$T = 150^\circ\text{C} + 273 = 423 \text{ K}$$

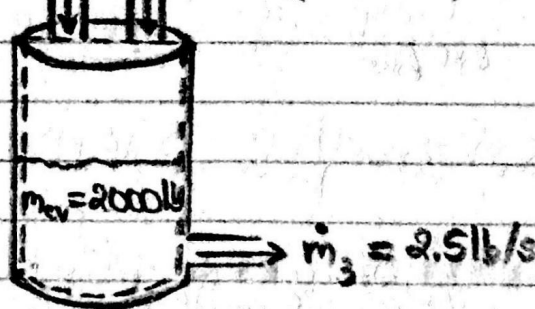
$$pv = RT \rightarrow v = \frac{RT}{PM} = \frac{(8.314 \text{ kJ/kmol K})(423 \text{ K})}{(200 \text{ kPa})(28.97 \text{ kg/kmol})} \times \frac{1000 \text{ Nm}}{1 \text{ kJ}} \times \frac{1 \text{ kPa}}{1000 \text{ N/m}^2}$$

$$\rightarrow v = 0.607 \text{ m}^3/\text{kg}$$

$$\dot{m}_e = \frac{A_e V_e}{v} \rightarrow \text{volumetric flow rate} = A_e V_e = \dot{m}_e v = (11.5 \text{ kg/s})(0.607 \text{ m}^3/\text{kg}) = 6.98 \text{ m}^3/\text{s}$$

2.) Sketch

$$\dot{m}_1 = 0.8 \text{ lb/s} \quad \dot{m}_2 = 1.2 \text{ lb/s}$$



Engineering Model

1. The system is a control volume

Analysis

$$\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = \dot{m}_1 + \dot{m}_2 - \dot{m}_3 = 0.8 \text{ lb/s} + 1.2 \text{ lb/s} - 2.5 \text{ lb/s} = -0.5 \text{ lb/s}$$

$$m_{cv,f} = m_{cv,i} + \frac{dm_{cv}}{dt} \Delta t = 2000 \text{ lb} + (-0.5 \text{ lb/s})(1 \text{ hr}) \left(\frac{3600 \text{ s}}{1 \text{ hr}} \right) = 200 \text{ lb}$$

1 Question 1 15 / 15

✓ - 0 pts Correct

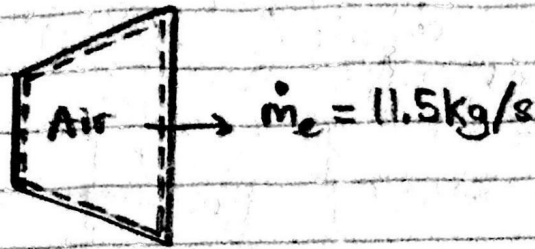
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CHE 201 - HW6

1.) Sketch

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$$T = 150^\circ\text{C}$$



Engineering Model

1. The system is a control volume
2. Ideal gas behavior

Analysis

$$T = 150^\circ\text{C} + 273 = 423 \text{ K}$$

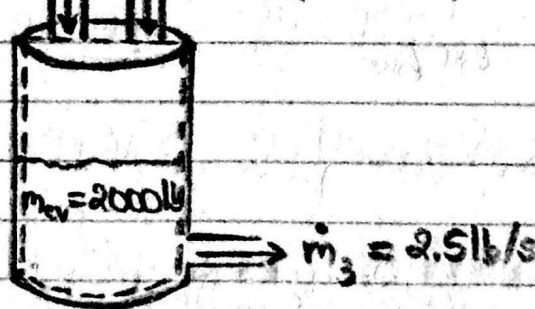
$$pv = RT \rightarrow v = \frac{RT}{PM} = \frac{(8.314 \text{ kJ/kmol K})(423 \text{ K})}{(200 \text{ kPa})(28.97 \text{ kg/kmol})} \times \frac{1000 \text{ Nm}}{1 \text{ kJ}} \times \frac{1 \text{ kPa}}{1000 \text{ N/m}^2}$$

$$\rightarrow v = 0.607 \text{ m}^3/\text{kg}$$

$$\dot{m}_e = \frac{A_e V_e}{v} \rightarrow \text{volumetric flow rate} = A_e V_e = \dot{m}_e v = (11.5 \text{ kg/s})(0.607 \text{ m}^3/\text{kg}) = 6.98 \text{ m}^3/\text{s}$$

2.) Sketch

$$\dot{m}_1 = 0.8 \text{ lb/s} \quad \dot{m}_2 = 1.2 \text{ lb/s}$$



Engineering Model

1. The system is a control volume

Analysis

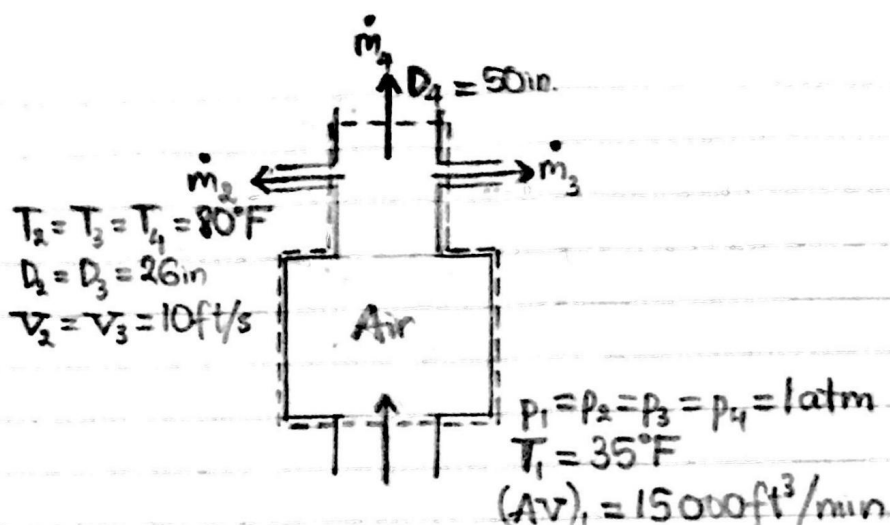
$$\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = \dot{m}_1 + \dot{m}_2 - \dot{m}_3 = 0.8 \text{ lb/s} + 1.2 \text{ lb/s} - 2.5 \text{ lb/s} = -0.5 \text{ lb/s}$$

$$m_{cv,f} = m_{cv,i} + \frac{dm_{cv}}{dt} \Delta t = 2000 \text{ lb} + (-0.5 \text{ lb/s})(1 \text{ hr}) \left(\frac{3600 \text{ s}}{1 \text{ hr}} \right) = 200 \text{ lb}$$

2 Question 2 15 / 15

✓ - 0 pts Correct

3.) Sketch



Engineering Model

1. The system is a control volume
2. Ideal gas behavior
3. Steady state

Analysis

(a) $T_1 = 35^\circ\text{F} + 459.67 = 494.67^\circ\text{R}$

$p_1 v_1 = RT_1 \rightarrow v_1 = \frac{RT_1}{p_1 M} = \frac{(0.7302\text{ ft}^3\cdot\text{atm}/\text{lbmol}\cdot^\circ\text{R})(494.67^\circ\text{R})}{(1\text{ atm})(28.97\text{ lb}/\text{lbmol})} = 12.47\text{ ft}^3/\text{lb}$

$\dot{m}_1 = \frac{(AV)_1}{v_1} = \frac{15000\text{ ft}^3/\text{min}}{12.47\text{ ft}^3/\text{lb}} \times \frac{1\text{ min}}{60\text{ s}} = 20.05\text{ lb/s}$

(b) $(AV)_2 = (AV)_3 = A_2 v_2 = \frac{\pi D_2^2}{4} v_2 = \frac{\pi (26\text{ in})^2}{4} \times \left(\frac{1\text{ ft}}{12\text{ in}}\right)^2 \times (10\text{ ft/s}) \times \frac{60\text{ s}}{1\text{ min}} = 2212\text{ ft}^3/\text{min}$

(c) $T_2 = T_3 = T_4 = 80^\circ\text{F} + 459.67 = 539.67^\circ\text{R}$

$\rightarrow v_2 = v_3 = v_4 = \frac{RT_2}{p_2 M} = \frac{(0.7302\text{ ft}^3\cdot\text{atm}/\text{lbmol}\cdot^\circ\text{R})(539.67^\circ\text{R})}{(1\text{ atm})(28.97\text{ lb}/\text{lbmol})} = 13.60\text{ ft}^3/\text{lb}$

$\frac{dm_{cv}}{dt} = 0 \rightarrow \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 - \dot{m}_3 - \dot{m}_4 = 0$
 $\rightarrow \dot{m}_4 = \dot{m}_1 - \dot{m}_2 - \dot{m}_3 = \dot{m}_1 - 2\dot{m}_2 = (20.05\text{ lb/s}) - \left(\frac{2212\text{ ft}^3/\text{min}}{13.60\text{ ft}^3/\text{lb}}\right) \times \frac{1\text{ min}}{60\text{ s}} \times 2 = 14.63\text{ lb/s}$

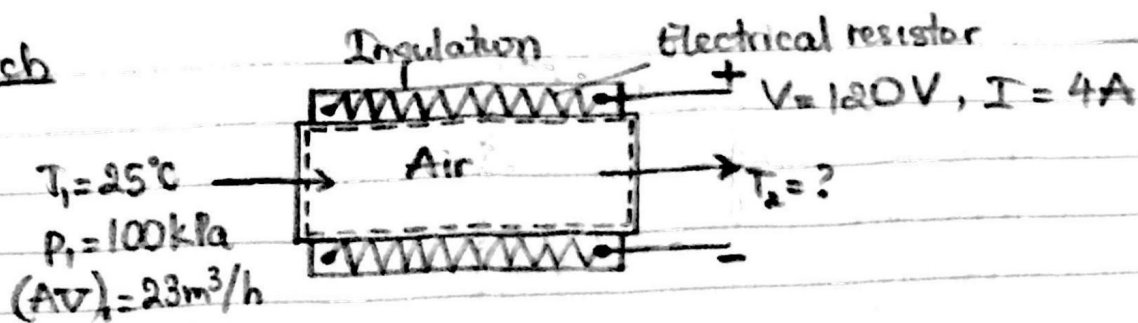
$\dot{m}_4 = \frac{A_4 v_4}{v_4} \rightarrow v_4 = \frac{\dot{m}_4 v_4}{A_4} = \frac{4\dot{m}_4 v_4}{\pi D_4^2} = \frac{4(14.63\text{ lb/s})(13.60\text{ ft}^3/\text{lb})}{\pi (50\text{ in})^2} \times \left(\frac{12\text{ in}}{1\text{ ft}}\right)^2 = 14.58\text{ ft/s}$

3 Question 3 15 / 15

✓ - 0 pts Correct

- 2.5 pts a) Wrong calculation
- 2.5 pts a) Wrong equation
- 2.5 pts b) Wrong calculation
- 2.5 pts b) Wrong equation
- 2.5 pts c) Wrong calculation
- 2.5 pts c) Wrong equation

4.) Sketch



Engineering Model

1. The system is a control volume at steady state
2. Ideal gas behavior, $c_p = 1.005 kJ/kg \cdot K$
3. $\Delta KE = \Delta PE = Q_{cv} = 0$

Analysis

(a) $T_1 = 25^\circ C + 273 = 298 K$

$$P_1 v_1 = RT_1 \rightarrow v_1 = \frac{RT_1}{P_1 M} = \frac{(8.314 kJ/kmol \cdot K)(298 K)}{(100 kPa)(28.97 kg/kmol)} \times \frac{1 kPa}{1000 N/m^2} \times \frac{1000 Nm}{1 kJ} = 0.855 m^3/kg$$

$$\dot{m}_1 = \frac{(AV)_1}{v_1} = \frac{(23 m^3/h)}{0.855 m^3/kg} = 26.90 kg/h$$

(b) $\dot{m}_1 = \dot{m}_2 = \dot{m} = 26.90 kg/h$

$$\frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2) = 0 \rightarrow -\dot{W}_{cv} + \dot{m}(h_1 - h_2) = 0$$

$$h_1 - h_2 = c_p(T_1 - T_2)$$

$$\dot{W}_{cv} = -P_{electrical} = -VI = -(120V)(4A) = -480W \times \frac{1 kJ/s}{1000W} = -0.48 kJ/s$$

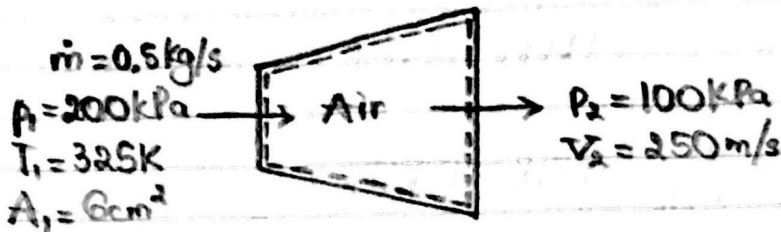
$$\rightarrow \dot{m}c_p(T_1 - T_2) = \dot{W} \rightarrow T_2 = T_1 - \frac{\dot{W}}{\dot{m}c_p} = (298K) - \frac{(-0.48 kJ/s)}{(26.9 kg/h)(1.005 kJ/kg \cdot K)} \times \frac{3600 s}{1 h}$$

$$\rightarrow T_2 = 362 K = 362 K - 273 = 89^\circ C$$

4 Question 4 20 / 20

✓ - 0 pts Correct

5.) Sketch



Engineering Model

1. The system is a control volume at steady state
2. $\dot{W}_b = \dot{Q}_w = \Delta PE = 0$
3. Ideal gas behavior with $c_p = 1.008 \text{ kJ/kg} \cdot \text{K}$

Analysis

$$(a) P_1 V_1 = RT_1 \rightarrow V_1 = \frac{RT_1}{P_1 M} = \frac{(8.314 \text{ kJ/kmol} \cdot \text{K})(325 \text{ K})}{(200 \text{ kPa})(28.97 \text{ kg/kmol})} \times \frac{1000 \text{ Nm}}{1 \text{ kJ}} \times \frac{1 \text{ kPa}}{1000 \text{ N/m}^2} = 0.466 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{A_1 V_1}{V_1} \rightarrow V_1 = \frac{\dot{m} V_1}{A_1} = \frac{(0.5 \text{ kg/s})(0.466 \text{ m}^3/\text{kg})}{6 \text{ cm}^2} \times \left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^2 = 388.3 \text{ m/s}$$

$$(b) \frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_w - \dot{W}_w + \dot{m}(h_1 - h_2 + \frac{1}{2}(V_1^2 - V_2^2)) = 0$$

$$\rightarrow \dot{m}(c_p(T_1 - T_2) + \frac{1}{2}(V_1^2 - V_2^2)) = 0 \rightarrow T_2 = T_1 + \frac{V_1^2 - V_2^2}{2c_p}$$

$$\rightarrow T_2 = (325 \text{ K}) + \frac{(388.3 \text{ m/s})^2 - (250 \text{ m/s})^2}{2(1.008 \text{ kJ/kg} \cdot \text{K})} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = 369 \text{ K}$$

$$(c) \frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \rightarrow V_2 = \frac{P_1 V_1 T_2}{P_2 T_1} = \frac{(200 \text{ kPa})(0.466 \text{ m}^3/\text{kg})(369 \text{ K})}{(100 \text{ kPa})(325 \text{ K})} = 1.058 \text{ m}^3/\text{kg}$$

$$\dot{m} = \frac{A_2 V_2}{V_2} \rightarrow A_2 = \frac{\dot{m} V_2}{V_2} = \frac{(0.5 \text{ kg/s})(1.058 \text{ m}^3/\text{kg})}{250 \text{ m/s}} = 2.116 \times 10^{-3} \text{ m}^2$$

$$\rightarrow A_2 = 2.116 \times 10^{-3} \text{ m}^2 \times \left(\frac{100 \text{ cm}}{1 \text{ m}}\right)^2 = 21.16 \text{ cm}^2$$

5 Question 5 20 / 20

✓ - 0 pts Correct

Current and Waves for Large-Scale Power Generation

The Viability of Waves for Large-Scale Power Generation in the Near Future

Making wave-power generations on a small scale is still considered as a very difficult work because there are many technological challenges—for example, high variability and unpredictability of wave energy—and industrial challenges. “Narrowing the broad variety of proposals to a few winning concepts does not automatically provide the industry with the optimal solution for large-scale wave energy production” [Reference 1]. However, for now, “the technology of small-scale units is maturing more rapidly” [Reference 1].

Technical and Economic Factors

Technical factors can include the inherent to the concept of current and wave energy and high variability and unpredictability of wave energy [Reference 1].

Economic factors can include the following:

- Spatial conflicts—for example, ports and harbors, pipeline areas, marine sanctuaries, military operations, oil and gas exploitation and mining, tourism, and so on [Reference 3].
- Interconnection to the power grid—for example, transmission requirements, integrating variable electric output, and shore landings [Reference 3].
- Capital and life-cycle costs—for example, installation, equipment, operation and maintenance, debris management, and so on [Reference 3].

Effects on the Ecosystem

“Ocean energy can contribute to a more sustainable energy supply, but it is not environmentally friendly per se” [Uihlein & Magagna]. In general, it will affect marine species, cause bottom disturbance, alter regional water movement, and change the composition of the area [Reference 3].

Benthic habitats will be affected by tidal current energy converters due to “the change of water flows and composition of substrate and sediment dynamics” [Uihlein & Magagna]. It affects “mortality of fish passing through the turbines and the collision risk of marine mammals with tidal stream farms” [Uihlein & Magagna]. One essential issue regarding the tidal energy converters is the noise disruption in the turbulent waters which will affect marine life.

Wave energy converters can also cause some potential harm to the sea life. One example is that it can “alter water column and sea bed habitats locally and by changes in the wave environment” [Uihlein & Magagna]. Other environmental impacts from this kind of converter may include “competition for space, noise and vibration, electromagnetic fields, disruption to biota and habitats, water quality change and possible pollution” [Uihlein & Magagna]. Although wave energy converters result in “a much lower collision risk compared to wind energy converters,” they tend to have a higher risk of “underwater collisions for diving birds” [Uihlein & Magagna].

References

1. "Can the Wave Energy Industry Produce on a Large Scale?" 31 Jan. 2013. URL: <https://www.energycentral.com/c/ec/can-wave-energy-industry-produce-large-scale>. Accessed 10 Mar. 2020.
2. Uihlein, Andreas and David's Magagna. "Wave and Tidal Current Energy—A Review of the Current State of Research Beyond Technology." *Renewable and Sustainable Energy Reviews*, vol. 58, May 2016, pp. 1070-1081. URL: <https://www.sciencedirect.com/science/article/pii/S1364032115016676#s0010>. Accessed 10 Mar. 2020.
3. "Wave, Tide, Ocean Current, In-Stream, Ocean Thermal (OTEC) Power: National Academy of Sciences 2013." 15 Jan. 2017. URL: <http://energyskeptic.com/2017/wave-tide-current-otec-power/>. Accessed 10 Mar. 2020.

6 Question 6 15 / 15

✓ - 0 pts Correct

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- 5 pts Not enough evaluation on technical factors
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